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RESEARCH

A longitudinal approach of trace element contamination of sediments in El Harrach river (North Central Algeria): characterisation and environmental quality

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Abstract A longitudinal study was conducted in 2020 and 2022 to evaluate trace element contamination in fluvial sediments of the El Harrach river within the catchment area, a semi-arid Mediterranean watershed, increasingly subject to anthropogenic pressures that affect Algiers Bay and the Mediterranean Sea. Forty-eight fluvial sediment samples were collected from eight sampling stations under contrasting hydrological conditions. Trace element concentrations were measured using X-ray fluorescence spectroscopy (XRFS), and geochemical indices were applied to assess the contamination level. The result revealed an overall moderate contamination, with significant enrichment of Cd, Co, Cu, Ni and Pb at downstream sampling stations, while As, Cr, Mo and Zn showed a low level of enrichment. Cd emerged as the most bioavailable element. PERMANOVA and PCA underscored the significant influence of seasonality and spatial variability, linking Ni with Ca, and associating As and Cr with iron/manganese oxides, suggesting predominantly natural geochemical controls. In contrast, Cd, Co, Cu, Zn and Pb were more strongly related to anthropogenic sources such as industrial, urban and agricultural activities. These findings eventually provide essential baseline data for managing sediment quality and preserving ecological integrity in Algiers Bay.

Keywords Fluvial sediments · Trace element · Contamination assessment · Bioavailability

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Introduction

Increasing concentrations of trace elements in watercourses have raised concerns regarding their effects on aquatic biodiversity, the quality of drinking water and public health. These trace elements can have harmful effects through bioaccumulation and biomagnification in the food chain (Kalala et al., 2016). They originate from both natural sources (e.g. weathered rocks, volcanic eruptions, sea spray and biomass combustion) and anthropogenic activities (e.g. industrial emissions, agriculture, traffic and wastewater) (Baril, 2014; Bliefert & Perraud, 2008; Pacyna, 1986). Evapotranspiration, although indirect, may increase trace

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element concentrations during droughts (Assaad, 2019). Once in the fluvial system, many trace elements adsorb onto suspended particles and accumulate in sediments, especially fine fractions (< 63 µm), where they may later be remobilised during hydrological events (Diop, 2014; Kalala et al., 2016). This dynamic influences their bioavailability and toxicity (Campbell & Tessier, 2018), thereby impacting the ecological integrity of aquatic systems (Burton et al., 1992; Dorea, 2003). The fluvial sediments are thus seen as both sinks and secondary sources of contamination (Adje et al., 2021), and their quality is a critical indicator of overall river health. Within environmental systems, the hydrosphere serves as a primary medium for the transfer of trace and major elements among the atmosphere, lithosphere and biosphere (Helali et al., 2009). Runoff and infiltration govern trace element mobility and influence surface and groundwater quality. Various contamination indices (EF, Igeo, SPI, ERI, PLI…) are used to assess trace element pollution and predict ecological risks (Adje et al., 2021; Djeddi et al., 2018). This is especially relevant in Mediterranean regions, where studies have shown significant accumulation of As, Cr, Ni, Pb and Zn in fluvial sediments (Ayari et al., 2021; El Houssainy, 2020; Makhoukh et al., 2011).

Situated within the Mediterranean basin, Algeria is subject to similar threats, with several rivers exhibiting trace element contamination (Benkaddour, 2018; Keddari et al., 2019; Sahli et al., 2014). The El Harrach river, which flows through the capital Algiers and discharges into the bay, is one of the most polluted rivers in Algeria. It is exposed to chronic anthropogenic pressure due to industrial discharge waste, urban runoff and untreated wastewater. While past research has investigated the river's hydrology (Med Boudiaf & Sarra, 2021) and water quality using bioindicators (Bouchelouche et al., 2019), few studies have addressed the trace element contamination of fluvial sediments in this basin (Benamar et al., 1999; Yoshida et al., 2006 and Atroune, 2013), particularly in connection with hydrological dynamics, land-use pressure and fluvial sediment transport. Consequently, there is no study of the effect of major human activity on fluvial sediments contamination, taking into account the influence of hydrological conditions, intensified erosion processes and the presence of dams affecting contamination transfer along the El Harrach river catchment. These factors, as highlighted by Taleb et al. (2004), have a direct influence on river water quality, which strongly recommends the knowledge and understanding of the transport and availability of trace elements and obviously is highly essential, both for preserving the integrity of ecosystems and for the protection of the human health (Nicolau, 2005).

This study aims to fill these knowledge gaps by addressing the following research question: What is the spatial distribution, extent and potential sources of trace element contamination in the fluvial sediments of the El Harrach river? We hypothesise that anthropogenic pressures, as well as hydrological variability and fluvial sediment dynamics, significantly influence both the concentration and spatial patterns of trace elements. This work is directly conducted with Algeria's ecological restoration initiatives, particularly the national program for the rehabilitation of urban watercourses, which includes the El Harrach river. By assessing contamination in fluvial sediments using geochemical and index-based approaches, this study provides valuable baseline data to guide local environmental decision-making and to support the design of pollution prevention and fluvial sediment management strategies in urban river systems. To answer this crucial question, our study set the following objectives:

- Assessing trace element contamination in fluvial sediments of the El Harrach river, identifying their distribution, origin and the impact of anthropogenic pressures using geochemical indices.
- Highlighting the impact of hydrological and environmental conditions on the behaviour and availability of trace elements.
- Assessing of the compliance of the El Harrach river sediments with international environmental quality standards.

Materials and methods

Study area and sampling strategy

Presentation of the study area

The El Harrach river catchment which covers around 1236.28 km^2, is located in the coastal region of Algiers and is drained by the El Harrach river over

67 km (Mimouni et al., 2009). It is divided into three parts: the Blideen Atlas (600 km^2), the Mitidja plain (500 km^2) and the Sahel (100 km^2) (Astite et al., 2015). The river's tributary network is dense (Brahim, 2015) and collects effluents from urban, agricultural and industrial sources. Key contributors include the Smar, Baba Ali, Terro and El Kerma rivers, which are mainly affected by domestic waste and discharges from various industrial zones such as chemical and metallurgical industries (Mezrag, 2013). Upstream, the catchment supports agriculture and tourism (notably around Hammam Melouane), while downstream, four industrial zones with over 470 units exert major pressure on water quality, with only 30% of this area connected to a wastewater treatment plant. The El Harrach river exhibits high seasonal flow variability, ranging from 0.01 to 3000 $m^3 s^{-1}$ (Aroua & Berezowska Azzag, 2009; Benghazi & Tobal, 2023) and is threatened by floods. Geologically, the basin lies in the Tellian chain and is mostly composed of Quaternary and Pliocene alluvium, underlain by marls, limestone and sandstones. In the plain, the blue marls are overlain by Astian deposits (upper Pliocene marine-coastal sediment) and Quaternary alluvium (Halimi, 1980; Mimouni, 2010; Zouakh, 1995). The climate is subhumid (Mebarki et al., 2013) with average rainfall ranging from 600 to 1200 mm and average yearly temperature ranging from 18 to 22 °C (2004–2014). The upper side of the catchment has a vegetation cover of 40–50%, including cypress, Aleppo pine, carob and lentisk bushes on the summits (Fergani, 2016). Arboriculture, market gardening and plasticulture are also present (Bouchelouche, 2015).

Station selection and location

Samples were collected within El Harrach river catchment, at eight sampling stations situated across the river, from its source to its mouth. These sampling stations were selected to assess the impact of various anthropic sources of trace elements on fluvial sediment quality, as well as the transfer of substances downstream to the Mediterranean Sea (see Fig. 1). The sampling stations are classified as follows: upstream, a reference sampling station (H1), followed by a sampling station along the river in the agricultural zone (H2). Sampling stations are then located at the discharge points of the tributaries of the El Harrach river (H3, H4, H5 and H7), then in urban and industrial areas. One of the sampling stations is located downstream before the wastewater treatment plant (H6), while the other is located at the interface where all discharges converge before flowing into coastal waters (H8).

Sampling protocol

In order to study the distribution of trace elements and assess fluvial sediment quality in the El Harrach river catchment, seasonal sampling was carried out in 2020–2022: three campaigns during low water periods (August 2020, August 2021 and October 2021) and three campaigns during high water periods (February 2021, December 2021 and February 2022) (Fig. 1). Superficial fluvial sediments were sampled with a plastic scoop, from the upper 3–5 cm of depth over an area of around 1 m^2 during six sampling campaigns at each sampling. Bedrock samples were also collected at each sampling station, from representative outcrop of the dominant drainage substrate. In the laboratory, fluvial sediment samples were air dried, pulverised by crushing, grinding and milling with a tungsten carbide mortar until an extremely fine powder was obtained. Rock samples were crushed and sieved < 75 µm, and the resulting powder was compressed under high pressure into solid, homogeneous pellets for analysis.

Analytical techniques

Mineralogical analysis

Mineralogical analysis was carried out by X-ray diffraction (XRD) conducted at the *Birine Nuclear Research Centre* (BNRC) of Algeria, using an X'Pert Pro MPD/Analytical diffractometer (goniometer radius 240 mm) equipped with a Pixel 1D detector to record the XRD spectra (Gravereau, 2011; Kubala-Kuku et al., 2013). These were recorded in GONIO (Theta-Theta) mode in the angular range from 10 to 75°2 ┐ with a step size of 0.026°2 ┐ and an acquisition time of 90 s. A copper anticathode was used as the source of radiation K~ = 0.1540598 nm. The radiation power used was 30 kV and 40 mA. The various crystalline phases contained in the samples were identified using records from an ICSD 2011 database (ICDD *International Centre for Data Diffraction*, USA) integrated with the XRD spectra processing

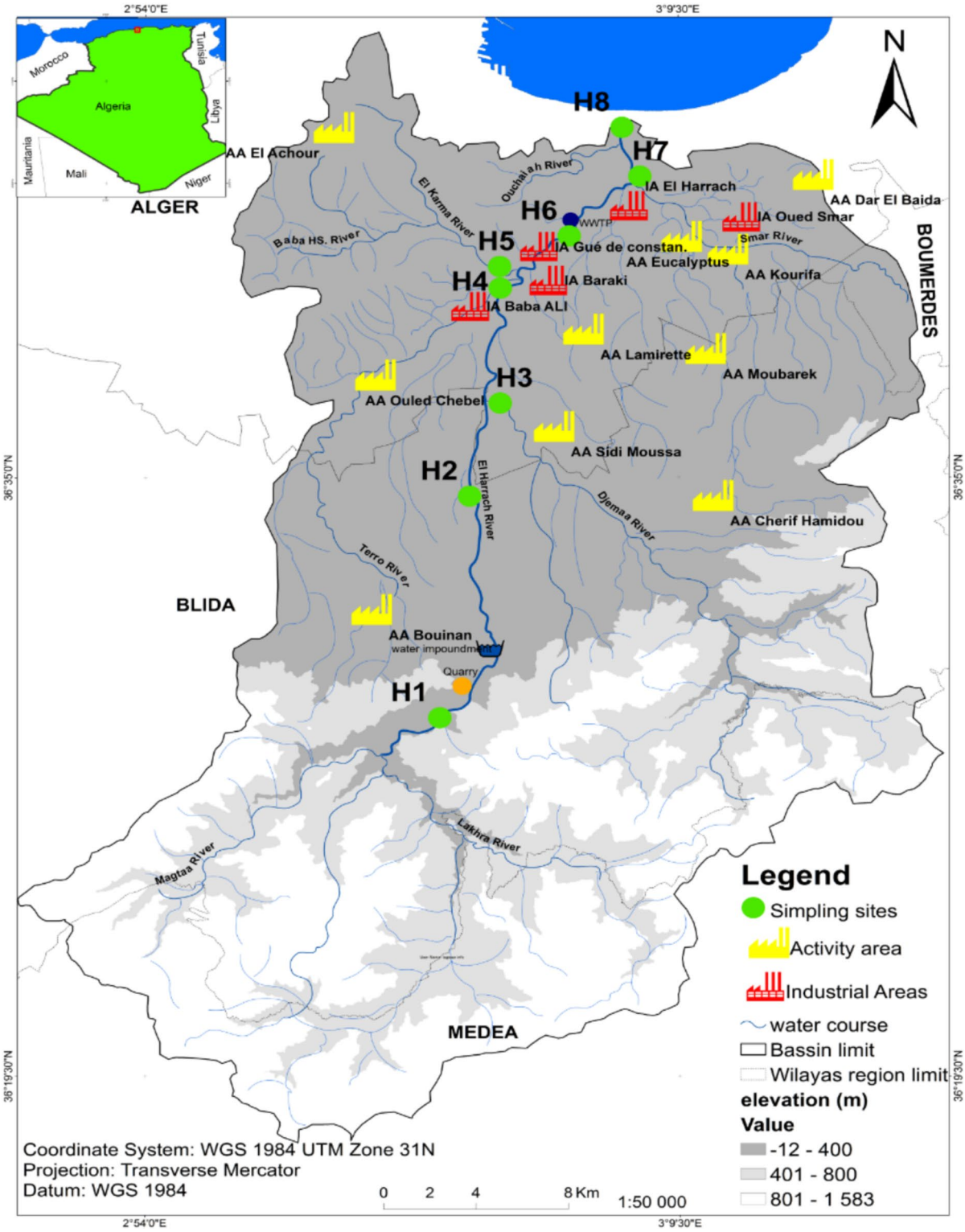

Fig. 1 Location map of sampling station within the El Harrach river catchment. Industrial area (IA): a zone dedicated to industrial establishments, production, processing, storage and logistics activities. Activity area (AA): a zone integrating various enterprises engaged in economic, industrial, commercial, service and office activities

software, High Score Plus (version 4.9 March 2020). The percentage of phases present in the samples analysed by XRD was determined using the Rietveld method. This method uses the pseudo-voigt function to refine several parameters of the XRD spectrum.

Particle size analysis

The particle size distribution of fluvial sediment samples was determined at the *Draria Nuclear Research Center* (DNRC) of Algeria, on a Malvern Mastersizer Hydro 2000 wet laser granulometer, under operating conditions (dispersant: distilled water and stirring speed: 2800 rpm). The results are displayed as a semi-logarithmic curve, illustrating the particle size distribution, where each value corresponds to the diameter of a sphere whose volume is equivalent to that of the particle. The particles measured are in the range 0.02–2000 µm. The concentration of the solution (suspension) should not be too high ($10 < \text{obscuration} < 20\%$).

Determination of major, minor and trace element concentrations

(a) In fluvial sediment and rock samples

Major (Al, Ca, Si and Fe), minor (Mn, P and S) and trace elements (Zn, Cr, Pb, Ni, Co, Cu, As, Mo and Cd) in fluvial sediment and rock samples were analysed using an Amptek type EDXRF device (EXP-1 XRF spectrometer kit), at the laboratory of the *Algiers Nuclear Research Center* (ANRC). X-ray fluorescence (XRF) is a powerful tool for monitoring the elemental composition and trace element content of soils, sediments, water and air particles (Benabdellah, 2005; Koltay, 1994; Malmqvist, 1994; RECORD, 2008; Van Grieken & Markowicz, 1993). It is an analytical technique that involves exciting the atoms in a sample using a radioactive source (X, γ) or an X-ray tube, and then measuring the energies and intensities of the characteristic X-rays emitted by each element. In this way, we determine the elements present in the sample and their concentrations through characteristic peaks for each element (Naili et al., 2016). Measurements were initially carried out by means of the regression method, with four standards supplied by the IAEA (*International Atomic Energy Agency*): PT-04 (clay), PT-05 (marine sediment), PT-09 (river clay) and PT-13 (clay). X-ray fluorescence spectra were analysed using Axil software (Analysis of X-ray Spectra by Iterative Least Squares), developed by the Physic's Department at the Seibersdorf Laboratory of the *International Atomic Energy Agency* (I.A.E.A.), in conjunction with multiple researchers at the University of Antwerp (Belgium) which can be used to extract the net intensities of predefined lines in qualitative analysis.

(b) In fluvial sediment extractable fraction

A simple extraction was carried out to determine the available fraction of trace elements in the fluvial sediments: One gram of fluvial sediments was leached with 10 ml of 0.05 mol EDTA at room temperature, for 1 h and filtered through a 0.22-µm pore size filter (Ghestem & Bermond, 1999). After dilution, the solution obtained was analysed using an energy-dispersive X-ray fluorescence (EDXRF) device manufactured by the *American company Amptek*, called the EXP-1 XRF spectrometer kit, at the *Algiers Nuclear Research Centre* (ANRC). Quantification of the trace elements present in our samples was carried out using the standard reference material (SRM) called NIST-1640a (NIST: the *National Institute of Standards and Technology*).

Data processing

Geochemical indices

To determine multielement contamination and to differentiate between natural sources and anthropogenic inputs in the fluvial sediments of the El Harrach river, several geochemical indices were applied including the enrichment factor (EF), the geoaccumulation index (Igeo), the ecological risk index (ERI), the sediment pollution index (SPI) and the pollution load index (PLI). In this study, geochemical background values were derived from the average trace element concentrations measured in the outcropping rocks of the El Harrach river basin, which better reflects the local geochemical setting. This choice ensures that the indices are representative of natural conditions specific to the study area. Each index was calculated following established

approaches (Hakanson, 1980; Müller, 1981; Rubio et al., 2000a, 2000b; Tomlinson et al., 1980), as detailed below.

(a) Enrichment factor (EF)

The enrichment factor (EF) quantifies the increase in concentration of a chemical element in a fluvial sediment relative to a reference material (Jean-Gael et al., 2019; Sterckeman et al., 2006; Wedepohl, 1995). It is widely used to discriminate between natural and anthropogenic sources, with Al or Fe commonly chosen as normalisers due to their natural abundance (Chester & Stoner, 1973; Rubio et al., 2000a, 2000b). In this study, Al was selected as the normalising element because of its low geochemical mobility, and association with aluminosilicate minerals (El Houssainy et al., 2020). The EF is calculated as:

$$EF = \frac{\left(\frac{Cx}{Al}\right)sample}{\left(\frac{Cx}{Al}\right)reference} \quad (1)$$

Cx: concentration of a given trace element and Al is the normalising element. EF values between 0.5 and 1.5 indicate a natural origin of trace elements, while those above 1.5 are attributed to anthropogenic inputs (Ouattara et al., 2021; Zhang et al., 2014). EF ⩽ 1 no enrichment; 1 < EF ⩽ 3 minor enrichment; 3 < EF ⩽ 5 moderate enrichment; 5 < EF ⩽ 10 moderately severe; 10 < EF ⩽ 25 severe enrichment; 25 < EF ⩽ 50 very severe enrichment (Konan et al., 2021).

(b) Geoaccumulation Index (Igeo)

The Igeo introduced by Müller (1981), is used to quantify the intensity of trace element contamination. It establishes a relationship between the measured concentration of trace elements in the fine sediment fraction and their local geochemical background concentration, including a corrective factor for lithological variations in these trace elements (Pitrat & Braga, 2012). It is defined by the equation:

$$Igeo = log2(C_n/1.5 \times B_n) \quad (2)$$

$Igeo$: geoaccumulation index
C_n: concentration measured in the sample
B_n: geochemical background; 1.5: background exaggeration factor defined a scale with six geoaccumulation index classes (Sahli et al., 2014). Igeo < 0 uncontaminated; 0 < Igeo < 1 uncontaminated to moderately contaminated; 1 < Igeo < 2 moderately contaminated; 2 < Igeo < 3 moderately contaminated to severely contaminated; 3 < Igeo < 4 severely contaminated; 4 < Igeo < 5 severely to extremely contaminated; 5 < Igeo extremely contaminated (Konan et al., 2021).

(c) Sediment Pollution Index (SPI)

Rubio et al. (2000) introduced the Sediment Pollution Index (SPI) to assess the degree of toxicity of fluvial sediments. The SPI is calculated according to the following formula:

$$SPI = \Sigma(CF_x \times W_x)/\Sigma W_x \quad (3)$$

CF: the contamination factor ($CF = C_{x\ Sample}/C_{x\ Bedrock}$ with C_x is the concentration of trace elements)
W: the toxicity weight of trace element x, with W_{Cr} and $W_{Zn} = 1$, $W_{Cd} = 300$, W_{Cu} and $W_{Ni} = 2$ and $W_{Pb} = 5$ (Rubio et al., 2000a, 2000b; Singh et al., 2002)

The sediment pollution index is based on four quality categories: 0⩽ SPI < 2 healthy sediment; 2 ⩽ SPI < 5 slightly polluted sediment; 5 ⩽ SPI < 10 moderately polluted sediment; 10 ⩽ SPI < 20 highly polluted sediment; SPI ⩾ 20 hazardous sediment (Konan et al., 2021).

(d) Ecological Risk Index (ERI)

The Ecological Risk Index (ERI) is a unique index used to represent the ecological risk of trace element pollution in fluvial sediments (Ahirvar et al., 2023). It was adopted by Hakanson (1980), taking into consideration pollutant concentration, pollutant type, degree of toxicity and the environmental response to that toxicity (Benkaddour, 2018). The ERI is calculated from the following equation:

$$ERI = \Sigma_i^n E_r^i \; ; E_r^i = T_f^i \times C_f^i = \frac{C_D^i}{C_R^i} \quad (4)$$

C_f^i: contamination coefficient
C_D^i: trace element concentration
C_R^i: trace element concentration in bedrock

T_f^i: trace element toxicity factor

The ERI qualifies fluvial sediments in several classes in order to assess the ecological risk that their contamination may generate: ERI < 150 low risk; 150 ⩽ ERI < 300 moderate risk; 300 ⩽ ERI < 600 considerable risk; ERI ⩾ 600 very high risk (Konan, 2021).

(e) Pollution Load Index (PLI)

The Pollution Load Index (PLI) serves to assess the overall degree of pollution in the fluvial sediments in the study area, based on the total concentration of all the trace elements investigated. For each site sampled, the pollution load index (PLI) can be calculated as the nth root of the product of n concentration factors (Ouattara et al., 2021). The expression of this index is

$$PLI = \sqrt{\left(CF_1 \times CF_2 \times CF_3 \times \ldots \ldots \times CF_n\right)} \quad (5)$$

CF: contamination factor ($CF = C_x$ *Sample*/C_x *Bedrock*)

n: number of trace elements. According to Tomlinson et al. (1980), this method identifies two levels of sample pollution: For PLI ≤ 1: absence of pollution and for PLI > 1: presence of pollution (Benabdelkader et al., 2018; Sougo et al., 2014).

Environmental quality

The average PEC quotient (QPECm) assesses the toxicity of multicontaminated fluvial sediments in terms of trace elements. The proposed toxicity threshold takes into account the effects of mixing trace elements (Mac Donald et al., 2000). The QPECm is represented by the following formula:

$$QPEC_m = \frac{\Sigma\left(C_i/PEC_i\right)}{n} \quad (6)$$

$QPEC_m$: average PEC quotient

C_i: measured concentration for a trace element *i*

PEC_i: Probable Effect Concentration for a trace element *i*

n: number of quotients calculated

The QPECm values were compared with the toxic risk threshold value (0.5) proposed by Mac Donald et al. (2000): no toxic fluvial sediments are characterised by a QPECm < 0.50 and toxic fluvial sediments by a QPECm > 0.50.

Statistical analysis

A permutation analysis of variance using the PAST 4.03 software (Anderson, 2001) was carried out to determine the significance of spatial and seasonal variability; a PERMANOVA with 9999 permutations was performed against the monthly sampling data. The similarity matrix was produced using Euclidean distances. Prior to statistical analysis, the data collections were normalised using Minitab 19 software, as the environmental variables were on different measurement scales. The PERMANOVA analysis compared the following parameters: Major, minor, and trace elements as well as fluvial sediment granulometry, during two hydrological periods (high water and low water), at eight sampling stations (H1, H2, H3, H4, H5, H6, H7 and H8) and the interaction of the two preceding factors. The two factors were defined as "fixed". A principal component analysis with the same values as those used in PERMANOVA was performed to avoid multiple hypothesis testing. It is designed to identify factors likely to influence the distribution of trace elements and their relationship with other environmental variables. Principal component analysis (PCA) and correlation matrix were performed using XLSTAT software, version 2023.2.0.14. The rest of the statistical processing was carried out with Microsoft Excel 2010.

Results

Physical, mineralogical and geochemical characteristics of fluvial sediments

Mineralogical phases

The mineralogical composition of fluvial sediments and bedrock revealed the presence of three main crystalline phases: dolomite, clay minerals and quartz. All sampling stations showed a dolomite and clay minerals phase, with similar percentages ranging from 40 to 59% in the fluvial sediment samples, except for sampling station H7 where dolomite was recorded at 92.4%. Quartz was found in much smaller quantities, not

exceeding 7%. Based on these observations, the fluvial sediment samples can be classified as having a mineralogical composition dominated by carbonates and silicates, specifically clay minerals and dolomite. Clay minerals clearly constitute primary components of the bedrock, accounting for over 90 % of its composition. Small quantities of dolomite and quartz are detected, but they do not reach more than 6%, with the exception of sampling station H5 where dolomite reaches 1.9%. This observation remains consistent across all sampling stations. According to the data, the bedrock is predominantly composed of clay minerals, with a relatively minor contribution from dolomite and quartz.

Particle size distribution

A granulometric study was carried out on the fraction below 2000 µm in the El Harrach river sediments in order to study the influence of texture on the concentrations of major, minor and trace elements in the fluvial sediments. It focused on two hydrological conditions: low water (LW) and high water (HW). The results obtained (Fig. 2) show that the clay content ranged between 0 and 0.90% for all sampling, whatever the hydrological conditions, with the exception of sampling station H4 which has a content of 2.44% during floods. The upstream sampling station (H1) and even those in the middle basin (H2, H3, H4, H5) are characterised by lower percentages of silt, fluctuating between 11 and 47%. However, the sampling stations downstream (H6, H7, H8) have higher percentages, approaching 60%. Most of the samples taken at the sampling stations show a predominance of sand, with the proportion varying from 30% downstream to 88% upstream. The percentage of silt is generally higher during low water periods than during high water periods. However, the percentage of sand increases considerably during periods of high water, especially at sampling station H1.

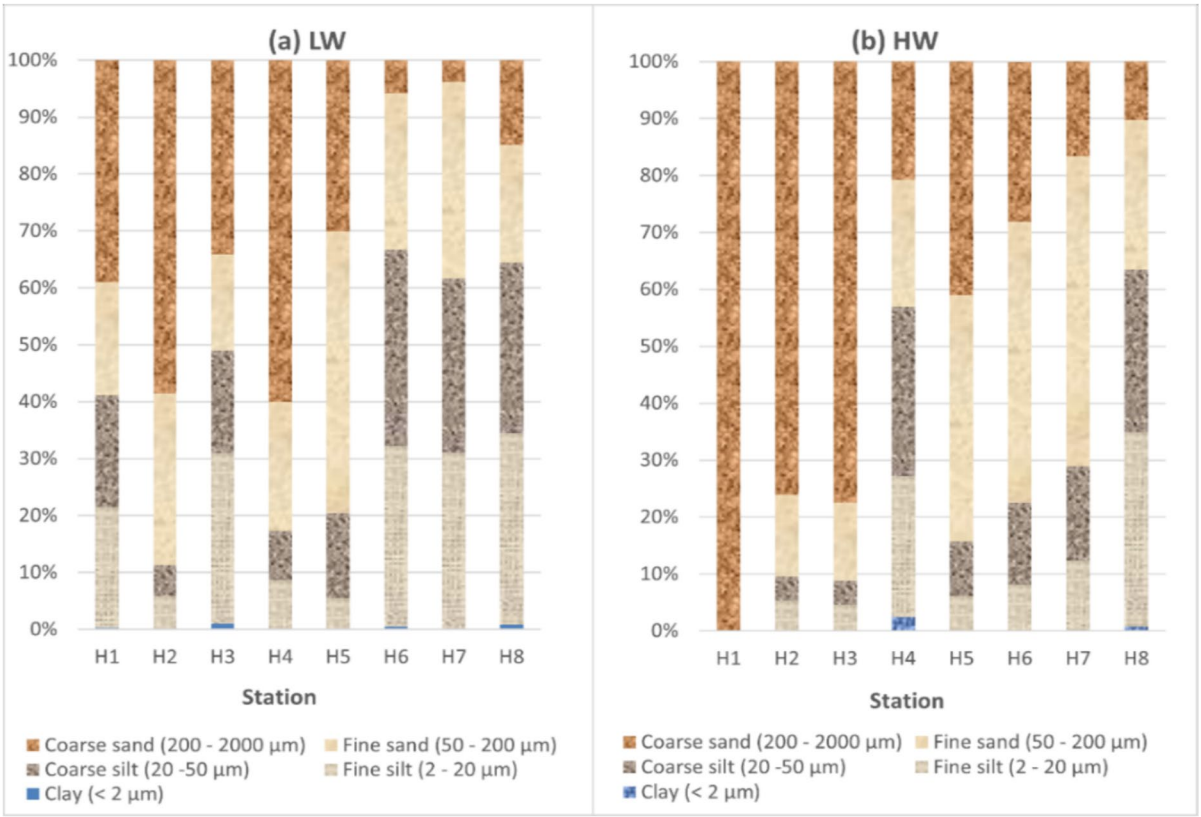

Fig. 2 Particle size distribution (< 2000 µm) in fluvial sediment samples collected along the El Harrach river under two different hydrological conditions: **a** LW: low water; **b** HW: high water

Distribution of trace element concentrations

Trace element content

The average concentrations of trace elements in the fluvial sediments followed the order of abundance: Zn > Cr > Pb > Ni > Co > Cu > As > Mo > Cd (Table 1). Pb (46.38 μg g^{-1}) and Zn (204.18 μg g^{-1}) showed the highest average concentrations, especially at downstream sampling stations (H7 and H6) while Cd and Mo were lowest (0.15 μg g^{-1} and 1.84 μg g^{-1}), particularly at sampling station H4. When compared to other Algerian rivers, the concentrations observed in the El Harrach river catchment are consistent with those reported for the Tafna and Chelif basins. In contrast, they are generally two to three times lower than those measured in North African rivers, such as the Sebou river in Morocco and the Rarai river in Tunisia. At the Mediterranean scale, trace element concentrations generally display patterns comparable to those reported for the Beirut river (Lebanon), the Rhône (France) and the Ebro (Spain). However, some trace elements deviate from this pattern. Cr (43.39 μg g^{-1}) is nearly ten times higher than in the Ebro river, while Pb (36.50 μg g^{-1}) is about twice the levels recorded in the Beirut river. Similarly, Pb, Cu and Co concentrations in the El Harrach river are two to three times higher than those in the Rhône river.

For major elements, Ca (91.42 mg g^{-1} and 87.95 mg g^{-1}) (at H1 and H8) and Si (86.94 mg g^{-1}) (at H3) showed the highest values.

Seasonal and spatial variations of trace element contaminations

Figure 3 shows the differences in trace element concentration distributions between upstream and downstream sampling stations under two hydrological conditions: high water (HW) and low water (LW). The disparities between upstream and downstream sampling stations are more visible, particularly for Zn, Pb and Co. Indeed, at H4, H5, H7 and H8, the concentrations of Zn and Pb were significantly higher than those measured at upstream sampling stations, during the two periods studied, while Co concentrations were higher upstream. However, it should be noted that these trace element levels decreased at sampling station H5 during the high water. Relative increases in Zn were significant, particularly at sampling station H7, regardless of the period considered. In the downstream section, concentrations tended to approach upstream values during periods of low water, with a slight increase observed during high water.

The results of the PERMANOVA analysis applied to the concentrations of major, minor, and trace elements, as well as percentage sediment grain size, showed significant effects for the sampling station factor ($F = 3.2892$, $p = 0.003$, permutation = 9999) and the season factor (PERMANOVA: season $F = 11.874$, $p = 0.0005$). No interaction between sampling stations and sampling periods was observed in these results and Fig. 3.

Assessment of trace element contamination

Trace element contamination in El Harrach river sediments was assessed using the enrichment factor (EF) under low (LW August 2021) and high (HW December 2021) water conditions, with local bedrock as reference materials (Fig. 4A/B). This analysis reveals differential contamination patterns. Co and Pb were the most enriched trace elements ranging from moderate ($5 < EF < 10$) to very high ($25 < EF < 50$) with stronger enrichment downstream, especially during the high water period. The enrichment of Cd (EF = 5.30), Cu (EF = 17.25) and Ni (EF = 11.96) was low to moderately severe and consistent across sampling stations and seasons. By comparison, As and Cr only slightly exceeded background levels, with the exception at sampling station H6. Mo and Zn generally showed low enrichment ($EF < 1.5$), except at sampling station H7 where Zn reached $EF = 9.33$ during high water conditions.

Several geochemical indices (Igeo: Eq. 2, SPI: Eq. 3, ERI: Eq. 4 and PLI: Eq. 5) were also applied to assess contamination levels (Table 2). Igeo values ($-4.59 \leqslant Igeo \leqslant 1.93$) indicated low contamination for most trace elements, in both hydrological conditions and across sites, with moderate levels for Cu (H5), Ni (H1) and Zn (H7). Co and Pb showed moderate contamination ($1 < Igeo < 2$) in 62–75% of samples during both seasons. SPI values ($SPI < 2$) classified all fluvial sediments as healthy. ERI values generally ranged between 300 and 600 indicating considerable ecological risks, with the highest risk at sampling station H7 (ERI = 626.96) and the lowest at sampling station H1 (ERI = 158.90). The PLI was < 1 at most sampling stations, suggesting limited overall

Table 1 Concentrations of major, minor and trace elements (mean and standard deviation) measured in the fluvial sediment samples collected along the El Harrach river during all campaigns as well as in the bedrock of the El Harrach river catchment (HR), the upper continental crust (UCC: Wedepohl, 1995) and fluvial sediments from other rivers (a) Benabdelkader et al. (2018); (b) Ayari et al. (2021); (c) Ocampo-Duque et al. (2008); (d) Ferrand et al. (2012). (-): not determined

			Major and minor elements (mg g^{-1})							Trace elements (µg g^{-1})								
			Al	Ca	Si	Fe	Mn	P	S	Zn	Cr	Pb	Ni	Co	Cu	As	Mo	Cd
Sediment	H1	x̄	14.81	91.42	70.38	21.78	0.13	2.72	0.17	12.85	33.63	23.73	31.68	14.60	14.43	3.70	1.90	0.15
		σ	4.33	13.55	8.05	6.85	0.12	0.35	0.07	1.59	5.47	8.06	15.20	5.59	1.38	0.64	0.25	0.05
	H2	x̄	11.58	74.25	72.13	30.83	0.26	3.73	4.47	52.50	48.48	39.65	19.78	25.35	20.00	3.90	1.89	0.25
		σ	3.35	3.60	10.74	4.57	0.11	0.69	2.97	32.85	5.69	9.33	4.36	9.73	4.34	0.70	0.23	0.05
	H3	x̄	14.73	47.83	86.94	40.24	0.34	3.19	1.55	23.37	52.67	35.5	10.83	15.03	14.33	4.27	2.36	0.23
		σ	0.97	9.37	4.81	2.19	0.04	0.31	1.13	1.31	8.83	3.10	4.05	5.90	2.05	0.24	0.88	0.05
	H4	x̄	14.19	68.56	79.23	30.46	0.22	2.96	2.00	32.2	49.28	34.7	20.38	26.65	16.88	4.10	1.84	0.18
		σ	2.63	9.39	5.69	3.28	0.09	0.36	1.00	15.77	5.18	2.74	2.15	10.25	3.23	0.32	0.07	0.04
	H5	x̄	12.27	68.22	82.78	26.92	0.21	3.46	1.84	117.6	45.75	37.05	23.73	11.38	26.70	4.25	2.32	0.25
		σ	2.69	13.91	8.84	2.69	0.08	0.46	1.00	97.16	8.78	13.09	1.83	3.60	5.77	0.26	1.06	0.05
	H6	x̄	12.23	67.79	84.57	29.84	0.18	3.73	1.33	68.43	41.88	46.38	19.90	20.7	22.45	3.60	2.82	0.30
		σ	1.22	1.97	14.77	1.81	0.10	0.33	0.86	31.08	6.96	16.21	8.90	11.64	3.12	1.06	1.63	0.07
	H7	x̄	11.52	112.64	66.94	20.68	0.24	3.89	1.44	204.18	34.55	33.4	26.18	16.38	17.83	3.43	2.42	0.25
		σ	1.02	20.43	13.17	4.43	0.09	0.27	0.75	117.69	6.18	14.38	3.17	8.08	3.03	1.07	1.20	0.11
	H8	x̄	12.02	87.95	76.07	27.29	0.18	3.39	0.77	62.65	43.25	41.35	23.28	20.23	15.65	3.98	2.29	0.23
		σ	4.31	24.17	14.29	6.32	0.12	0.42	0.43	37.90	8.78	13.06	2.78	9.25	0.92	0.23	1.10	0.04
	Mean H	x̄	12.86	78.28	77.07	28.13	0.21	3.39	1.70	73.28	43.39	36.5	22.33	18.91	18.67	3.89	2.23	0.23
		σ	3.19	22.76	12.67	6.97	0.11	0.57	1.78	83.39	9.47	12.88	8.71	9.87	5.24	0.72	1.01	0.08
	Min. H		7.25	35.92	54.18	12.35	0.01	2.23	0.10	11.20	24.70	13.30	6.00	2.50	11.50	1.60	1.55	0.09
	Max. H		21.70	138.49	109.47	42.69	0.38	4.63	8.64	392.00	65.00	71.30	56.7	35.00	35.60	4.60	5.63	0.40
	Tafna (a)		30.72	113.97	-	17.63	0.33	0.58	0.74	74.31	40.83	43.37	16.63	6.61	17.51	5.13	-	0.19
	Rarai (b)		-	-	-	-	-	-	-	180.00	57.40	147.0	36.70	-	22.60	85.8	-	0.64
	Rhône(c)		-	-	-	-	-	-	-	61.30	44.60	14.60	25.10	6.30	9.96	-	-	0.08
	Ebro (d)		-	-	-	-	-	-	-	45.75	4.74	23.92	15.55	-	13.25	6.59	-	0.22
Bedrock	Mean HR	x̄	35.66	183.57	150.12	9.82	0.24	0.94	0.02	106.88	57.84	12.47	17.58	6.63	11.45	6.15	24.94	0.20
		σ	14.78	47.18	75.96	9.28	0.16	0.39	0.00	71.88	21.10	6.78	2.43	5.73	5.42	5.67	8.08	0.03
	UCC (e)		77.44	29.45	303.48	30.89	0.53	0.66	0.95	52.00	35.00	17.00	18.6	11.6	14.3	2.00	1.40	0.10

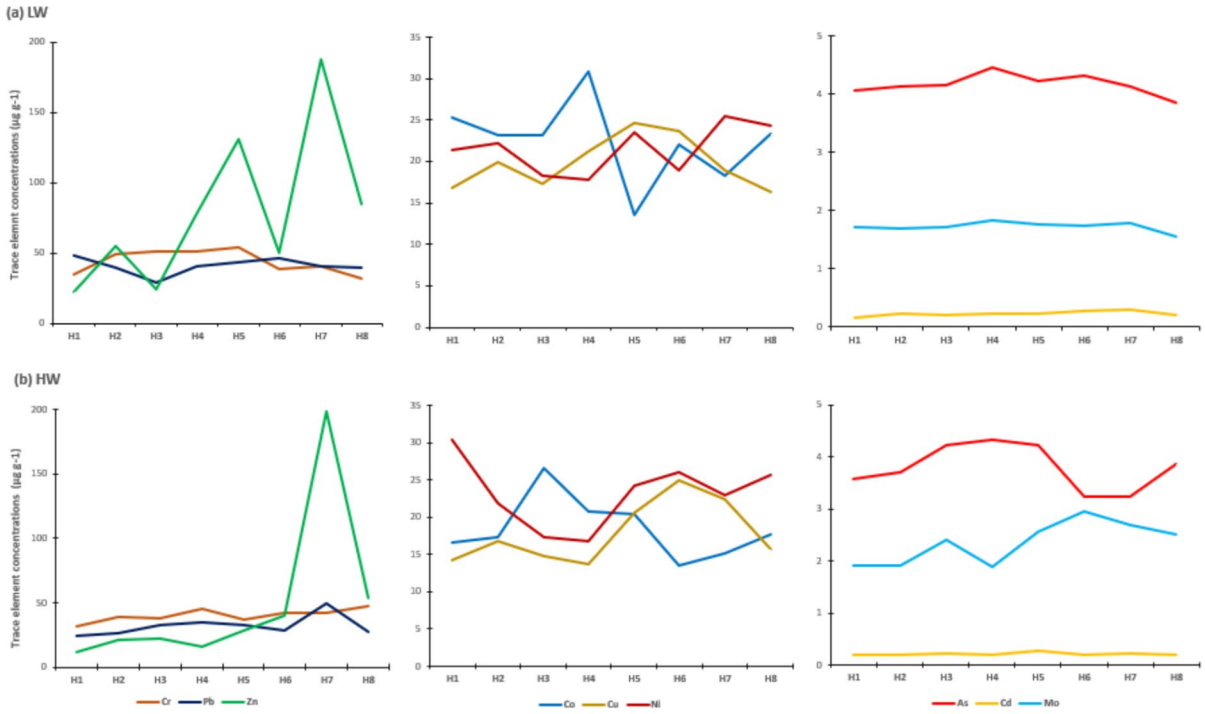

Fig. 3 Distribution of trace element concentrations in fluvial sediments at sampling stations in the El Harrach river catchment studied during two distinct hydrological conditions: **a** low water (LW) and **b** high water (HW)

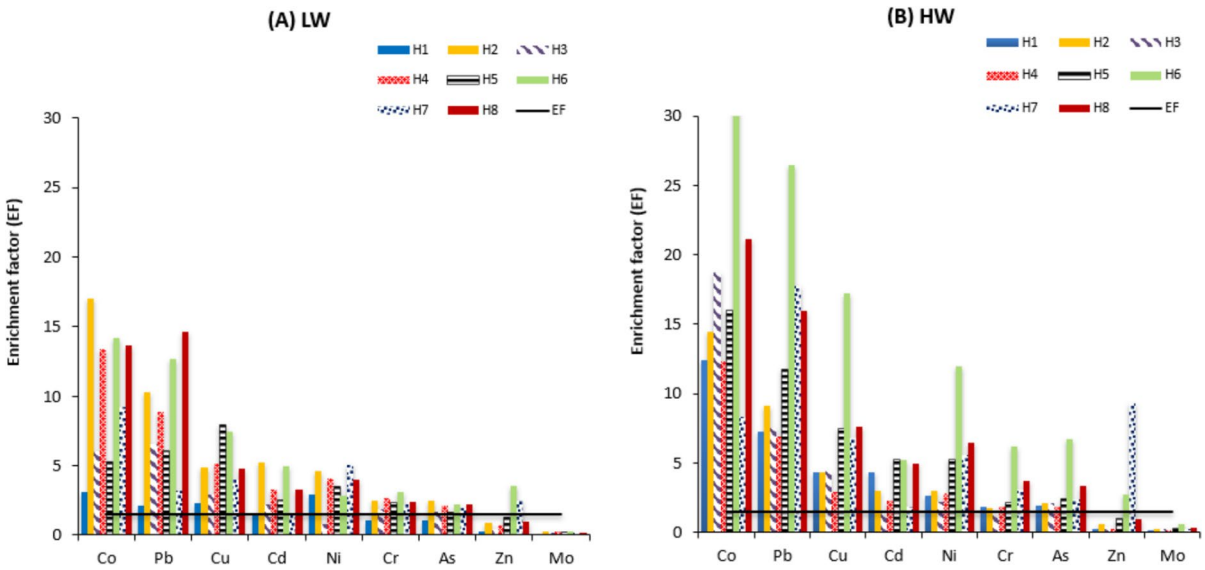

Fig. 4 Variation of the enrichment factors (EFs) for trace elements in the fluvial sediments across the different sampling stations under contrasting hydrological conditions: **A** low in August 2021, **B** floods in December 2021). The horizontal dashed line symbolizes the limit of natural variability (EF = 1.5)

Table 2 Contamination indexes (Igeo: geoaccumulation index, SPI: sedimentary pollution index, ERI: ecological risk index and the PLI: pollution level index) calculated for the fluvial sediment samples, collected during several sampling campaigns (low waters period: August 2020; August 2021 and high waters period: February 2021; February 2022)

		Igeo Aug.20									Igeo Aug. 21									SPI Aug. 20	SPI Aug. 21	ERI Aug. 20	ERI Aug. 21	PLI Aug. 20	PLI Aug. 21
		Cr	Co	Ni	Cu	Zn	As	Mo	Cd	Pb	Cr	Co	Ni	Cu	Zn	As	Mo	Cd	Pb						
LW	H1	−1.18	1.27	−0.38	−0.12	−3.84	−1.17	−4.35	−1.61	0.93	−1.19	0.34	0.23	−0.12	−3.43	−1.17	−4.49	−0.61	−0.19	0.54	0.98	174.05	314.80	0.17	0.16
	H2	−0.76	1.71	−0.09	0.67	−0.57	−1.07	−4.46	−0.61	1.42	−1.09	1.68	−0.19	−0.11	−2.54	−1.1	−4.38	−0.03	0.96	1.03	1.46	331.11	469.63	1.68	0.59
	H3	−0.42	1.03	−1.31	−0.08	−2.88	−1.17	−4.49	−0.61	1.04	−0.85	0.86	−2.14	−0.18	−2.78	−1.00	−4.37	−0.61	0.98	1.00	1.00	321.92	321.14	0.29	0.19
	H4	−0.61	1.82	−0.57	0.02	−2.08	−1.10	−4.31	−0.61	1.01	−0.92	1.43	−0.29	0.05	−2.87	−1.24	−4.44	−0.61	0.83	1.01	1.00	323.02	320.69	0.67	0.38
	H5	−0.94	0.34	−0.36	0.67	0.80	−1.21	−4.52	−0.03	0.98	−0.73	0.47	−0.12	1.05	−1.55	−1.21	−4.40	−0.61	0.66	1.48	1.01	473.55	323.21	1.42	0.67
	H6	−1.30	0.64	−1.34	0.11	−2.01	−1.21	−4.40	−0.61	1.93	−0.70	1.49	−0.82	0.56	−0.52	−1.17	−4.39	−0.03	1.33	1.04	1.48	334.96	475.41	0.36	1.43
	H7	−1.29	1.02	0.01	−0.24	−0.37	−1.24	−4.54	−0.61	0.57	−1.02	1.02	0.16	−0.19	−0.87	−1.24	−4.50	−1.61	−0.49	0.99	0.52	319.25	166.47	0.61	0.30
	H8	−1.27	1.01	0.03	−0.20	−0.34	−1.21	−4.54	−0.61	0.58	−1.04	1.46	−0.31	−0.05	−2.30	−1.17	−4.59	−0.61	1.56	1.00	1.03	319.71	329.55	0.64	0.54
		Igeo Feb. 21									Igeo Feb. 22									SPI Feb. 21	SPI Feb. 22	ERI Feb. 21	ERI Feb. 22	PLI Feb. 21	PLI Feb. 22
HW	H1	−1.81	0.33	1.1	−0.41	−3.53	−1.21	−4.4	−1.76	0.50	−1.37	−0.10	−0.50	−0.38	−3.81	−1.83	−4.02	−0.61	−0.20	0.50	0.97	158.90	310.36	0.13	0.07
	H2	−0.63	1.52	−0.93	0.08	−1.81	−1.14	−4.39	−0.03	1.31	−0.92	−0.19	−0.59	0.11	−2.49	−1.77	−4.03	−0.61	0.47	1.47	0.98	473.29	315.59	0.77	0.19
	H3	-	-	-	-	-	-	-	-	-	−0.95	−0.55	−0.73	−0.58	−2.68	−1.17	−3.38	−0.03	0.74	-	1.45	-	465.72	0.00	0.24
	H4	−0.77	1.82	−0.47	0.26	−1.54	−1.04	−4.32	−1.61	0.98	−1.00	−0.01	−0.18	−0.55	−3.53	−1.32	−4.31	−0.61	0.73	0.55	0.99	176.46	318.49	0.61	0.15
	H5	−0.69	0.48	−0.09	0.47	−0.64	−1.07	−4.43	−0.61	1.64	−1.46	−0.93	−0.07	0.22	−2.30	−1.00	−3.17	−0.03	0.33	1.04	1.45	333.84	465.87	1.12	0.34
	H6	−1.18	1.81	−0.38	0.58	−0.94	−1.04	−4.16	0.39	1.11	−1.10	−1.14	0.37	0.24	−2.07	−2.36	−2.73	−0.03	0.50	1.93	1.44	620.72	463.75	1.65	0.35
	H7	−1.33	1.19	−0.33	0.28	1.29	−1.10	−4.26	0.39	1.47	−1.77	−1.99	0.07	0.28	0.41	−2.53	−3.06	−0.03	1.13	1.95	1.46	626.96	469.95	2.69	0.46
	H8	−1.23	1.51	−0.40	−0.06	−2.08	−1.14	−4.37	−0.61	1.52	−0.58	−0.93	−0.09	−0.24	−1.6	−1.36	−3.16	−0.03	0.61	1.02	1.45	329.02	465.89	0.58	0.48

pollution, except in February 2021 when it exceeded 1 at sampling stations H6 (1.65) and H7 (2.69).

Relationships between major, minor, and trace element concentrations

(a) Spearman's correlation

The correlation matrix revealed significant inter-element relationships that provide insights into the geochemical behaviour of major, minor, and trace elements in El Harrach river sediments (Table 3). Strong positive associations were observed between Cr, Fe and Mn ($0.70 < \rho \leq 0.85$, $p < 0.01$), reflecting their common occurrence in iron–manganese oxides and hydroxides. Ni was the only trace element significantly correlated with Ca ($\rho = 0.76$, $p < 0.01$), suggesting its association with carbonate phases. Moderate correlations were also identified between As and Cr with Fe and Si, as well as between As and Al, pointing to their affinity with aluminosilicate minerals.

For minor and trace elements, Cd and Zn showed the strongest relationship with P ($0.83 < \rho \leq 0.90$, $p < 0.01$), while Cu, Mo and Pb were also well correlated with P, indicating possible enrichment linked to organic matter and phosphate inputs (Jiao et al., 2012). Cr displayed a significant association with S ($\rho = 0.73$, $p < 0.01$), suggesting co-precipitation with sulphide minerals under reducing conditions. Among trace elements, the most pronounced interactions were observed between As and Cr ($\rho = 0.81$, $p < 0.01$), as well as among Cd, Cu and Zn, which likely share common anthropogenic sources (e.g. industrial effluents, urban discharges). Moderate correlations were also observed between Cd and Pb, and between Mo and Zn.

In contrast, strong negative correlations were recorded, mainly between Al/P and Zn/P, Ca/Si, Fe/Ni and Cr/Ni. These negative relationships may indicate contrasting geochemical affinities and highlight different source contributions (natural vs. anthropogenic). Overall, the correlation patterns suggest that some trace elements (Cd, Cu, Zn, Pb) are mainly derived from anthropogenic inputs, while others (Cr, Ni, As) reflect a mixed origin involving both natural geochemical matrices and human activities.

(b) Principal component

In support of this initial examination of the data, it is interesting to have a more synthetic overview (ACP: Fig. 5), including the entirety of the data collected from the El Harrach river and its tributaries, i.e. forty-eight fluvial sediment samples (eight sampling stations and six campaigns) and 16 chemical element variables as a function of two hydrological conditions: high water and low water periods. Examination of the results reveals that during high water, the three main components explain 79.6% of the total variation, with the first two accounting for 66.27%. Al (71%), As (70%), Co (88%), Cr (50%), Fe (76%) and Mn (78%) are strongly positively correlated with the first axis, while Ca (66%), Cd (78%), Cu (69%), Ni (65%), Pb (55%) and Zn (75%) are negatively correlated. The second axis is positively associated with Mo (67%), P (70%) and Si (76%), while Cr (55%) and S (64%) are positively correlated with the third axis. During low water, the relationships between the variables change. Ca and Ni, which were strongly correlated with axis 1 during high water, now conflict with Al, As, Cd, Cr, Fe, Mo, Si and Pb. This indicates a change in the associations between these elements during the low water period. In addition, elements such as Mn, P, S and Zn show opposite associations to Al compared to those observed during high water.

Bioavailability of trace elements and the environmental quality of fluvial sediments

The toxicity of trace elements is not only linked to their total concentration, but above all to their availability to living organisms (Coulibaly et al., 2013). Due to the inherent damage to living organisms caused by trace elements, the quantity of bioavailable trace elements is considered to be a key factor in the fate of aquatic ecosystems (Adechina et al., 2018). The results in Table 4 present data from a simple extraction of fluvial sediments from the El Harrach river over two distinct hydrological periods. The bioavailability of most of the trace elements was considerably reduced (< 5%), while Cu and Mo were found to be bioavailable at a rate of < 20% under both hydrological conditions. On the other hand, Cd was potentially available at a rate of between 20 and 90%, particularly during high water periods.

Table 3 Spearman's correlation matrix for major, minor and trace elements of the El Harrach river sediments. underlined values: significant correlation with $p < 0.05$ ($N = 48$)

	Al	Si	P	S	Ca	Mn	Fe	Cr	Co	Ni	Cu	Zn	As	Mo	Cd	Pb
Al	1															
Si	0.405	1														
P	−0.886	−0.168	1													
S	−0.214	0.190	0.228	1												
Ca	−0.381	−0.976	0.108	−0.333	1											
Mn	−0.252	0.180	0.307	0.731	−0.287	1										
Fe	0.190	0.643	−0.168	0.548	−0.714	0.563	1									
Cr	0.167	0.619	−0.180	0.738	−0.667	0.707	0.857	1								
Co	−0.357	−0.024	0.156	0.405	−0.024	0.204	0.476	0.310	1							
Ni	0.024	−0.667	−0.108	−0.524	0.738	−0.623	−0.952	−0.786	−0.476	1						
Cu	−0.500	0.048	0.659	0.381	−0.119	−0.072	−0.190	−0.119	0.095	0.048	1					
Zn	−0.762	−0.119	0.838	0.095	0.143	0.036	−0.452	−0.262	−0.048	0.214	0.738	1				
As	0.500	0.643	−0.503	0.405	−0.619	0.347	0.571	0.810	−0.190	−0.429	−0.238	−0.357	1			
Mo	−0.238	0.286	0.551	−0.381	−0.238	0.000	−0.238	−0.310	−0.357	−0.024	0.214	0.548	−0.310	1		
Cd	−0.663	0.196	0.902	0.233	−0.258	0.191	0.000	−0.074	0.098	−0.270	0.786	0.773	−0.331	0.638	1	
Pb	−0.381	0.500	0.407	0.095	−0.476	−0.060	0.405	0.214	0.333	−0.548	0.452	0.333	0.071	0.262	0.626	1

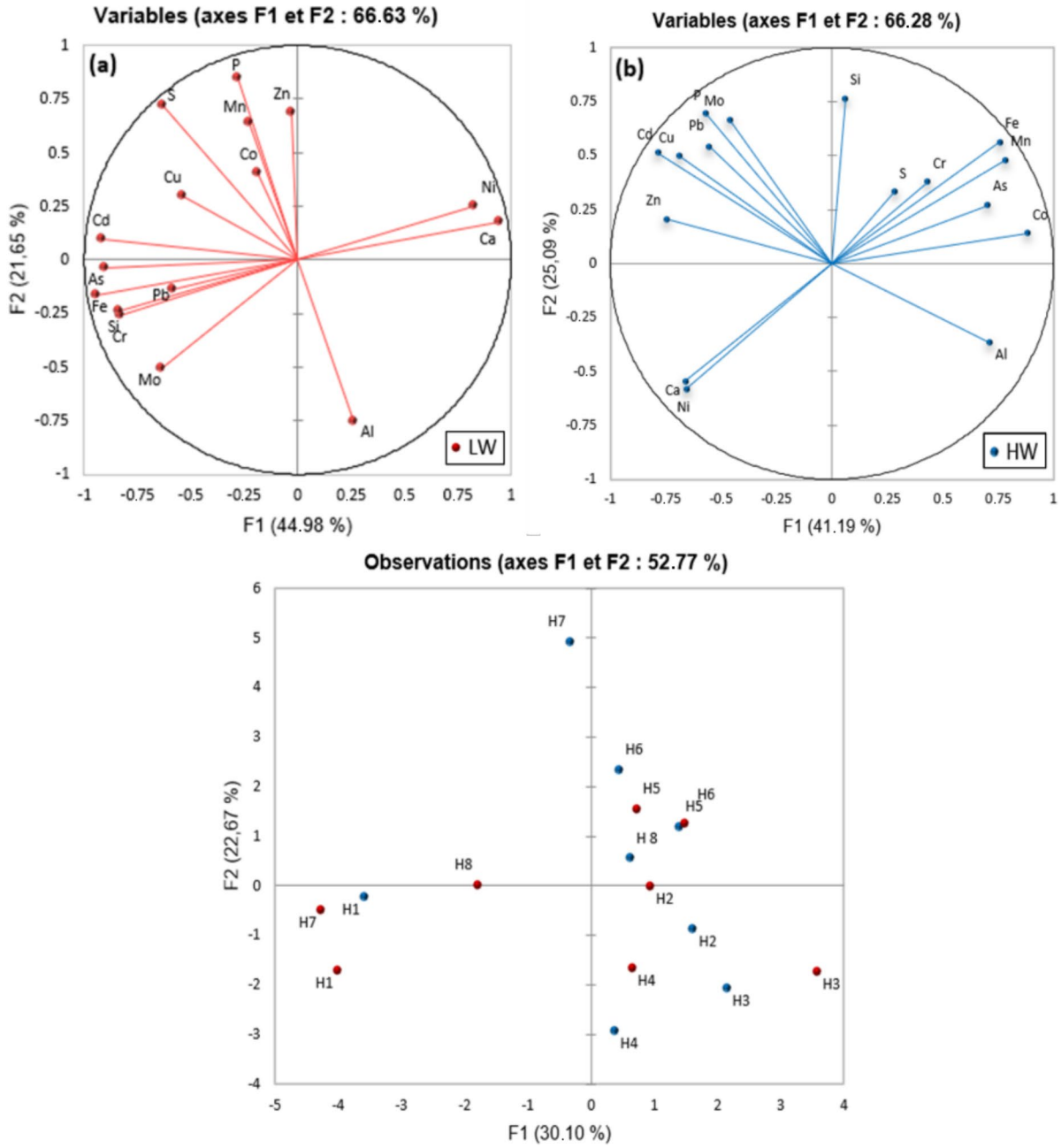

Fig. 5 Graphical approach to the PCA of major, minor and trace elements measured in fluvial sediments according to the plan (F1xF2). **a** Factor map of variables in low water conditions. **b** Factor map of variables in high water conditions. **c** Factor map of sampling stations in the two hydrological conditions

In addition, in order to assess the environmental quality of the fluvial sediments, the trace element concentrations in the fluvial sediments of El Harrach river were compared with the recently developed American SQGs (Sediment Quality Guidelines), based on a database of biological and ecological effects that could be caused by certain concentrations of contaminants in the fluvial

Table 4 Mean concentrations of trace elements in the available fraction (μg g^{-1}) and their corresponding proportions of the total concentration (%), in fluvial sediments from all sampling stations within the El Harrach river catchment, obtained through EDTA extraction, for low water (LW) and high water (HW) conditions

		Cr		Co		Ni		Cu		Zn		As		Mo		Cd		Pb	
		μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%	μg g^{-1}	%
LW	H1	14.6	0.40	14.3	1.19	24.5	0.83	73.0	4.85	26.0	1.83	3.69	0.95	5.9	3.72	6.44	37.92	4.91	0.31
	H2	15.3	0.39	21.1	0.70	23.6	1.07	78.1	5.16	25.6	0.97	4.15	1.01	11.8	6.89	10.51	33.96	5.49	0.16
	H3	14.9	0.33	18.6	1.08	24.7	4.32	79.1	5.46	28.9	1.30	3.46	0.79	10.3	5.98	9.86	42.54	2.51	0.07
	H4	15.0	0.34	17.9	0.70	24.8	1.21	75.0	4.42	34.4	1.65	5.63	1.52	15.9	9.73	13.43	70.50	5.03	0.16
	H5	14.6	0.29	15.0	1.14	24.8	1.07	84.2	2.48	88.1	1.69	3.76	0.99	13.6	8.07	7.47	45.90	3.42	0.12
	H6	14.8	0.29	16.8	0.63	24.5	1.73	73.3	3.03	74.3	0.70	9.97	2.55	32.6	19.14	8.00	32.56	14.53	0.32
	H7	14.7	0.36	16.5	0.86	25.3	0.90	77.7	5.40	40.8	0.49	5.87	1.58	4.20	2.65	3.44	30.85	7.02	0.55
	H8	14.7	0.36	16.6	0.64	25.1	1.24	72.4	4.58	24.0	0.77	6.65	1.70	10.0	6.79	8.15	41.36	6.12	0.12
HW	H1	14.7	0.43	16.6	0.63	25.2	1.69	76.6	4.76	33.5	3.62	11.82	3.03	28.3	17.78	6.71	20.91	12.95	0.44
	H2	14.5	0.47	18.5	0.63	25.0	1.52	78.0	5.09	22.3	1.19	5.86	1.43	15.3	9.01	9.75	41.66	5.41	0.15
	H3	14.6	0.52	18.4	0.47	25.2	2.07	77.6	4.88	20.2	1.54	3.46	0.84	13.7	7.76	9.62	70.10	4.25	0.14
	H4	14.8	0.34	15.6	0.47	24.5	1.23	79.7	5.93	20.4	1.98	3.04	0.68	10.0	5.27	10.30	44.78	3.31	0.09
	H5	14.7	0.45	17.6	0.63	25.9	1.05	83.6	3.67	23.2	0.79	4.53	1.13	15.5	8.85	8.04	24.78	4.41	0.11
	H6	14.7	0.46	18.5	0.87	25.3	1.33	78.2	4.39	28.4	1.07	8.65	2.33	29.7	20.92	11.83	86.96	5.33	0.18
	H7	14.8	0.32	18.2	1.27	25.8	1.02	77.2	3.84	31.0	0.12	2.21	0.58	16.7	9.80	7.59	55.45	4.14	0.07
	H8	15.0	0.37	16.9	0.64	25.4	1.18	80.8	4.88	18.1	0.94	10.45	2.68	19.7	12.36	18.76	90.42	3.17	0.08

sediments. The TEC (Threshold Effect Concentration) and PEC (Probable Effect Concentration) were established by MacDonald et al. (2000), are widely used to assess the quality of marine and fluvial sediments. The TEC identifies the concentrations of contaminants below which living organisms in fluvial sediments are not affected, unlike the PEC, which identifies the concentrations of contaminants above which harmful effects on living organisms in fluvial sediments are observed (Ingersoll et al., 1996; MacDonald, 1997).

Analysis of the data presented in Table 5 shows that most trace element concentrations in fluvial sediments are either below or slightly above environmental toxicity thresholds (TEC), suggesting levels of potential concern. More specifically, sampling stations H3, H5 and H6 show Cr levels slightly above the TEC in low water, while sampling station H7 records this exceedance in high water. As for Ni, sampling station H1 shows high concentrations at low water levels, while sampling stations H5 and H7 show levels above the TEC, particularly during floods. In addition, sampling station H7 merits particular attention, as it reached a Pb level of 59.8 μg g^{-1}, while the Zn concentration there considerably exceeded the TEC, reaching 121 μg g^{-1}, with significant attenuation to 270 μg g^{-1}. However, the concentrations measured in relation to the PEC were relatively low, with quotients (QPECm) of less than 0.5 for all the sampling stations.

Discussion

Level, extent and source of trace element contamination

Spatial differences in trace element concentrations reflect varying anthropogenic pressures and lithological backgrounds (Adje et al., 2021; Jean-Gael et al., 2019). The outcomes of contamination indices (EF, Igeo, SPI) for Cd, Co, Cu, Ni and Pb reveal a consistent pattern, and low values for As, Cr, Mo and Zn (EF < 1.5 and EF < 1, respectively) further indicate minimal enrichment, in agreement with findings from other Mediterranean rivers such as Boumerzoug (Sahli et al., 2014), Sebou (El Mrissani et al., 2021) and the Gulf of Lion (Roussiez et al., 2006). Conversely, sampling station H7 exhibited high Zn levels, likely linked to landfill leachates and discharges from nearby industries (Aroua & Berezowska-Azzag, 2009; Pitrat & Braga, 2012).

The detection of trace elements in the upstream section (H1) of El Harrach river indicated that their anthropogenic origin is primarily linked to thermal power plants and urban runoff from nearby villages of Hammam-Melouane (Morsi, A., 2016). It should also be noted that Ni recorded the highest value upstream and always correlates significantly with Ca, which explains its origin at this sampling station (H1). In fact, an increase in trace elements content in fluvial sediment does not necessarily mean anthropogenic

Table 5 Mean concentrations of trace elements (μg g^{-1}) in fluvial sediments from all sampling stations within the El Harrach river catchment, during the two contrasting hydrological periods (LW: low water; HW: high water), compared with the threshold values for fluvial sediment quality: Threshold Effect Concentration (TEC, below which adverse effects of benthic organisms are unlikely), Probable Effect Concentration (PEC, above which harmful effects are expected) and mean PEC quotient (QPECm) (MacDonald, 1997)

	H1		H2		H3		H4		H5		H6		H7		H8		USA	
	LW	HW	LW	HW	LW	HW	LW	HW	LW	HW	LW	HW	LW	HW	LW	HW	TEC	PEC
Cr	37.9	36	40.8	32.7	48.2	29.6	45.8	45.1	52.3	34.3	53.5	33.8	42.8	48.8	42.2	42.9	43.4	111
Co	12.6	27.9	31.8	30.9	18	40.9	26.7	34.6	13.8	29.5	28	22.4	20.2	15	27.4	27.7	-	-
Ni	31	15.7	23.1	17.2	6	12.8	21.6	20.9	24.3	25.9	14.9	19.9	29.5	26.5	21.3	22.5	22.27	48.6
Cu	15.8	16.9	15.9	16.1	15.2	16.7	17.8	14.1	35.6	23.9	25.4	18.7	15.1	21.1	16.6	17.4	31.6	149
Zn	14.9	9.7	27.6	19.7	23.3	13.8	21.9	10.8	54.9	30.7	112	27.8	87.5	270	32.6	20.2	121	459
As	4.1	4.1	4.3	4.3	4.6	4.3	3.9	4.7	4	4.2	4.1	3.9	3.9	4	4.1	4.1	9.79	33
Mo	1.67	1.67	1.8	1.78	1.81	1.85	1.72	2	1.77	1.84	1.79	1.49	1.65	1.79	1.55	1.67	-	-
Cd	0.2	0.3	0.3	0.2	0.2	0.1	0.2	0.2	0.2	0.3	0.3	0.1	0.1	0.1	0.2	0.2	0.99	4.98
Pb	16.4	30.6	36.3	36.9	36.9	30.9	33.2	36.6	29.5	40.6	47.2	31.2	13.3	59.8	55	39.5	35.8	128
QPECm	0.19		0.20		0.16		0.20		0.24		0.22		0.27		0.22			

enrichment, and in some cases, it may come from the high natural geochemical background (Adamou et al., 2022).

In contrast to upstream inputs, in the middle section (H2 and H3: sampling stations located in an area characterised by high agricultural and domestic activity), trace element contamination appears tied to agricultural runoff and fertiliser use, especially for Cd, Cu, P and Zn (Khous, 2020; Otero et al., 2005; Roussiez et al., 2006). H3 was additionally affected by tannery effluents and municipal waste leachates (Paul, 2017; Rahman et al., 2022). Notably, Co has been significantly enriched likely due to the many mining operations that have been developed (Belaroui et al., 2023) and to craft activities in this region, such as the manufacture of porcelain and ceramics using Co-based blue tint, noting that this has also been observed in Terro river sediments. Cr concentrations, in the fluvial sediments, are close to the natural geochemical background and the statistical correlation with the Fe/Mn pair, characterised as a natural leaching phenomenon of these trace elements present in the source rocks. Moreover, the significant enrichment by trace elements observed in sampling station H2 can be explained by the anthropogenic impact, which is reflected in the recreational use of the upstream parts of the river and the reshaping of the watercourse by straightening the channel (Hafiane et al., 2016), over the Douera dam to the west of the catchment area. In this respect, the effect of dilution may be diminished because during the period of high flow, large loads of suspended solids will be trapped behind this reservoir (Probst et al., 1999; El Mrissani et al., 2021). This limits the natural transfer of water, reduces the speed of flow and consequently increases the residence time of fluvial sediments and encourages their deposition (Fremion et al., 2016).

Moving further downstream, industrial activities become the dominant sources of contamination, particularly at sampling stations H6, H7 and H8. These stations showed significant enrichment of Cd, Co, Cu, Ni and Pb (see Table 1 and Fig. 4) mainly due to industrial runoff and urban activities during high flow (Manoj & Padhy, 2014; Ouattara et al., 2021; Yoshida et al., 2006). Pb enrichment, particularly at sampling station H6 (EF > 25), is likely linked to emissions from battery manufacturing, fuel combustion and former storage sites (Gaujous, 1995; Khaled-Khodja et al., 2016). In addition, this area was potentially impacted by service stations used by vehicles, and the presence of old fuel storage banks suggests another local origin, causing this contaminant to escape into the river. The local source of Pb in sampling station H7 is probably due to discharges from telephone cable manufacturing units. The enrichment of fluvial sediments by Co, particularly in the Terro river sediments (H4), is probably linked to the metallurgy industries and earthenware manufacturing units in this region. Cu shows slightly elevated concentrations in the El Karma river sediments (H5), mainly linked to discharges from electric cable manufacturing industries as a local source. The highest contamination occurred at sampling station H6, while contaminant levels decreased towards sampling station H8, likely due to the retention capacity of the Baraki wastewater treatment plant. This retention effect contributed to a moderate reduction of contaminant loads in the river (El Mrissani et al., 2021) outside. Another factor is dredging at the mouth of the river for clean-up and decontamination, which has been recognised by local stakeholders as effective in removing fluvial sediment deposits that trap contaminants (Mihoubi et al., 2012; Ocampo-Duque et al., 2008). It can result in the remobilisation of contaminants that have accumulated over the years, leading to their transfer to the Mediterranean Sea. When released into the water column, those contaminants can exhibit significant toxicity. Thus, there is a risk that trace elements are being remobilised from fluvial sediments, both deposited and reworked, could enter benthic and pelagic networks, potentially disrupting the food chain (Fichet, D., 1997; Hamzeh, M., 2012 and Diop, C., 2014).

Overall, the geochemical indices converge with the multivariate analyses. PERMANOVA and PCA highlighted two main groups of influences on trace element distribution: anthropogenic inputs (industrial, urban and agricultural) and natural lithogenic contributions which reflect the geochemical background and can generate anomalies but do not constitute contamination.

Interestingly, the distribution of trace elements highlights the geomorphological characteristics of the study area, which is a large, elongated watershed with a relatively pronounced relief, stretching from the Blidean Atlas range to the Sahel region. In mountainous areas, trace elements are easily mobilised by soil erosion and transported downstream by runoff, while the slower flow promotes their deposition, in

the coastal plain (Musy & Higy, 2004). Moreover, the variation in trace element concentrations in the El Harrach river sediments is influenced by two main factors. As Colinet (2003) points out, on the one hand, natural factors such as climate, parent material and relief, and on the other hand, anthropogenic factors including land use, proximity to industrial activities (Jean-Gael et al., 2019), and the artificial development of the watercourse. The latter has been channelled and modified to avoid obstacles and dangerous bends at the ends. These activities, aimed at improving flood flow conditions, have permanently altered the physical composition of the river (Aroua & Berezowska-Azzag, 2009), resulting in direct densification and dilution effects linked to flow (Keddari et al., 2019).

Bioavailability and environmental quality of sediments

According to the results (Table 4), the availability of trace elements in El Harrach river sediments was assessed based on their chemical forms rather than total concentrations, which can be misleading indicators of toxicity (Adechina et al., 2018; Zhou, 2009). The mobility sequence observed (Cd > Mo > Cu > Zn > As > Ni > Co > Cr > Pb) mirrors trends in other Mediterranean basins (Tafna, Sebou), notably the high bioavailability of Cd. Despite their high total concentrations, Co and Pb exhibited low bioavailability (< 5%), confirming that total content does not necessarily indicate ecological risk (Leleyter et al., 2012; Adamou et al., 2022). Cd, however, showed high mobility (up to 70% at H3), likely due to phosphate fertilisers rich in Cd (Keddari et al., 2019) and the formation of soluble Cd–phosphate complexes (Sadiq, 1992), particularly under competitive binding conditions with other cations (Elmayel et al., 2020). Co and Pb are mainly bound to Fe oxides and silicates, Ni to carbonates and Zn to phosphates, reducing their availability (Yao & Kouassi, 2015). Organic matter also plays a role in adsorbing trace elements and limiting release. Located downstream, the sampling stations H6 and H8 pose a significant threat of contamination of living organisms with Cd, even though concentrations are considered to be relatively low.

To evaluate ecological risks, trace element concentrations were compared to TEC and PEC thresholds (MacDonald et al., 2000) (Table 5). Most values were below TEC, indicating no expected toxicity, except for Pb and Zn at sampling station H7, which exceeded TEC, suggesting a potential risk for sensitive species. However, all concentrations remained below PEC, meaning sediments are generally non-toxic. This aligns with the results of SPI, PLI and low QPECm values.

In summary, spatial patterns highlight both natural and anthropogenic controls, but certain trace elements pose disproportionate risks. Co and Pb emerged as the most critical contaminants, showing consistent enrichment across several sampling stations, particularly H2 and H7, while SPI and PLI indicate generally low overall pollution loads and local hotspots associated with industrial and urban inputs present considerable ecological risks. This aligns with previous findings in the Tafna river (Benabdelkader et al., 2018) and Gascogne river (N'guessan et al., 2009) and underscores the buffering effect of carbonates, which reduce trace element toxicity (Tessier et al., 1982). Notably, Cd exhibits a particularly high level of bioavailability. These findings underline the importance of focusing on Pb, Co and Cd management strategies in the El Harrach catchment. Nevertheless, index-based assessments have limitations. Total trace element concentrations do not predict bioavailability or toxicity (Castañé et al., 2003; Witters, 1998). Speciation, influenced by pH, organic matter and redox conditions, determines the true environmental risk (Souza et al., 2015). Additionally, the combined (cocktail) effects of multiple contaminants (e.g. hydrocarbons, pesticides, pharmaceuticals) are not captured by traditional indices (Adje et al., 2021; Keddari et al., 2019).

Conclusion

By combining mineralogical, geochemical and statistical analyses, this study provided new insights into the contamination status of El Harrach river sediments and the mobilisation pathways of trace elements. The assessment of fluvial sediment samples' contamination using geochemical indices reveals that the indices (EF and Igeo) indicate a more or less significant enrichment in downstream sampling stations, particularly for Cd, Co, Cu, Ni and Pb. This is likely attributable to anthropogenic inputs from industrial and urban sources, as well as the application of phosphate fertilizers. Nevertheless, a low level

of enrichment was observed for As, Cr, Mo and Zn, in El Harrach river sediments, due to a lithogenic origin. The results of SPI with PLI show that these fluvial sediments are not very polluted, indicating a low level of contamination for the trace elements studied. This is reflected in the assessment of their bioavailability (by EDTA extractability) and the environmental quality of the fluvial sediments, so Cd appeared to be the most bioavailable trace elements, exhibiting high mobility and greater ecological risk for aquatic organisms. At sampling station H7, its availability exceeded that of Zn, indicating poorer sediment quality.

Principal component analysis revealed a close link between certain trace elements such as As and Cr and iron oxides. An association was observed between Ca and Ni. Elements such as Cd, Co, Cu, Pb and Zn are mainly derived from anthropogenic activities. They are distributed in fluvial sediments according to their affinity for various phases such as oxides, silicates and phosphates. This speciation is influenced by physicochemical characteristics of fluvial sediments.

The upstream sampling station (H1), which is generally characterised by a coarse sediment structure, is the least contaminated by these trace elements. The upstream sampling station (H2) and the downstream sampling stations (H6, H7 and H8) are identified as the most affected by trace elements. Moreover, the infrastructure such as upstream reservoirs and the Baraki WWTP were shown to influence trace element distribution, partly mitigating downstream contamination.

Overall, our results strongly suggest that there has been a significant improvement in the quality of El Harrach river sediments, resulting from a decrease in average concentrations of trace elements. This improvement is directly attributable to the control of industrial discharges and advances in urban wastewater collection and treatment.

Although the methods employed in this study yielded significant insights, the integration of complementary techniques, such as isotopic analysis and sequential extraction, would substantially deepen our understanding of contamination mechanisms. The methodological constraints of our research limited the assessment of ecotoxicological risks, particularly in terms of sampling depth and the range of contaminants examined. As a result, an in-depth chronological analysis using sediment cores and the analysis of organic micropollutants and environmental factors is necessary. Such an approach, combined with accurate mapping of contaminated areas and the development of sediment quality standards specific to the Algerian context, would represent a major step forward for the sustainable management of this essential ecosystem.

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Authors' contributions A. TALEB, N. BELAIDI and A. BENABDELKADER selected sampling sites, which were validated by N. BENMOUSSA for here PhD work. N. BENMOUSSA conducted the sediment sampling. N. BENMOUSSA and I. TOUMERT participated in the preparation of sediments and the analysis of trace elements. N. BENMOUSSA, A. TALEB, N. BELAIDI and A. BENABDELKADER collaborated on the data analysis. N. BENMOUSSA wrote the main manuscript text . A. TALEB, N. BELAIDI and A. BENABDELKADER reviewed and annotated the article.

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Declarations

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